

business models, are the most common type of MUBs in projects before deployment and also the second hardest to find in manual auditing, due to the need of understanding the most complex parts of contracts, namely, the business logics. On the other hand, their impact can be devastating. An example would be the Uranium Finance Exploit [14]. Due to two extra zeros in an interest calculation, the contract was exploited for \$57 million USD. The bug survived multiple rounds of manual auditing (by experts).

In this paper, we develop a type-checking tool for accounting bugs in smart contracts. Our insight is that *although accounting bugs reside in complex business logic and seemingly lack an application-agnostic oracle, many manifest themselves as abstract type violations. Abstract type inference* is a technique that can be traced back to the 70's in the last century [40]. It aims to abstract higher level semantic information such as physical units (e.g., seconds and meters) than those denoted by primitive types in programming languages such as integers and strings. As such, type systems can be enhanced to check a much richer set of properties [16, 31, 33, 40–42, 50, 56, 59, 61], such as physical unit consistency in robotic systems. We further observe that *although smart contracts have sophisticated business models, their basic operations are still analogous to those in a simple bank system*, such as deposit, withdraw, exchange, and loan. We hence devise an abstract type system based on these operations that can infer and check abstract types. In particular, we model and infer three facets of each variable, which are: *token unit* indicating the kind of currency denoted by the variable (analogous to USD in real life), *scaling factor* that denotes how much the variable has been scaled in order to simulate floating point computation that is not supported in smart contract programming languages, and *financial meaning*, e.g., if the variable denotes an interest or a debt. With the rich types, we can check a large set of properties that shall be uniformly true for various business models, using type rules. For instance, values of different token units cannot be added or subtracted together, similar to how lengths of meters and inches cannot be added together; amounts scaled by different factors should not be compared; interest should not be subtracted from debt but rather adds to it. More details can be found in section 3. To use our tool, the user annotates a few global variables and function parameters whose abstract types cannot be inferred from code (likely because they are implicit in pre-conditions and not reflected in implementation). The annotations are limited (see section 4) and usually clear from project description and even variable names. Then, our technique automatically infers the abstract types for other variables and performs type checking. Our system is flow-sensitive, context-sensitive and field-sensitive.

Our contributions are summarized as follows.

- We devise a novel abstract type system for smart contracts based on two observations: (1) many accounting bugs manifest themselves as abstract type errors; and (2) operations in smart contracts with complex business models can be abstracted to a few primitives and their combinations.
- Our type system models three aspects, token unit, scaling factor, and financial meaning, providing a good coverage for common type errors (according to our experiments).
- We implement a prototype `ScType` based on Slither [26]. We evaluate the system on 29 contracts from [78] that have 57 reported accounting bugs. Among these bugs, we find that

```

1  contract Pools{
2  ...
3  function addLiquidity(address base, address token, address
    member) external returns(uint liquidity) {
4      uint addedBase = getAddedAmount(base, ...);
5      uint addedToken = getAddedAmount(token, ...);
6      liquidity = calcLiquidityUnits( addedBase, totalBase,
        addedToken, totalToken, totalLiquidity);
7      liquidity[...][member] += liquidity;
8      totalBase += addedBase; ...
9  }
10 function calcLiquidityUnits(uint b, uint B, uint t, uint T,
    uint P) external view returns (uint){
11     uint part1 = (t * B);
12     uint part2 = (T * b);
13     uint part3 = (T * B) * 2;
14     uint _units = (((P * part1) + part2) / part3);
15     return (_units) / one;
16 }

```

Figure 1: Buggy Code from *Vader* [15]

33 of them are type errors and our method detects 29 of the type errors (i.e., 87.9% recall). Our tool reports 14 false positives due to its lack of path-sensitivity. The remaining 57-33=24 bugs are mostly due to pure math incorrectness and beyond type systems. We also apply our technique in a most recent audit contest organized by Code4Rena [2] for a very large project with more than ten thousand lines of code and find six zero-day accounting bugs in nine contracts that we check. Four can lead to direct fund loss. We have included our tool and benchmarks in the supplementary document. We will release the tool upon publication.

2 MOTIVATION

We use two real-world accounting bugs to explain the inadequacy of existing techniques and illustrate our method.

Example I (Lending Contract *Vader*). Figure 1 contains code detailing two functions included within the *Pools* contract from the *Vader* project [15]. They have been shortened for demonstrative purposes. *Vader* is a lending project that enables the pooling of funds and offers borrowing functionalities. Users can participate by purchasing shares from a liquidity pool and utilize these shares as collateral to access borrowed funds, thereby increasing the pool's returns. Within the project, The Pools contract stores functions related the movement of liquidity in the pool.

The function `addLiquidity()` in Figure 1 converts a contract's recently added *base* currency and *token* currency into liquidity, and adds the liquidity to a certain member's account. In the contract, the base currency is the default currency of the *Vader* project, while the token currency is a special currency used by the liquidity pool, representing a share of the pool. The amount of recently added base currency is computed on line 4 as `addedBase`, and the amount of recently added token is computed on line 5 as `addedToken`. The equivalent amount of liquidity is calculated through the `calcLiquidityUnits()` function call on line 6, in which `totalBase`, `totalToken`, and `totalLiquidity` denote the total amount of base currency, total token currency, and total liquidity in the pool. The resulting liquidity is then added to the member account on line 7 and the total base currency is updated on line 8.

```

1  contract LibBalances{
2  ...
3  function applyTrade(Position position, Trade trade, uint256
    feeRate) internal pure returns ... {
4      int256 signedAmount = trade.amount;
5      int256 signedPrice = trade.price;
6      int256 quoteChange = signedAmount * signedPrice;
7      int256 fee = getFee(trade.amount, trade.price, feeRate);
8
9      int256 newQuote = 0;
10     int256 newBase = 0;
11     if (trade.side == LONG) {
12         newBase = position.base + signedAmount;
13         newQuote = position.quote - quoteChange + fee;
14     } else if (trade.side == SHORT) {
15         newBase = position.base - signedAmount;
16         newQuote = position.quote + quoteChange - fee;
17     }
18     ...
19 }

```

Figure 2: Buggy Code from Tracer [13]

The bug occurs within the function `calcLiquidityUnits()`. Associating the variables on line 6 to the formal arguments on line 10, we have that `b` is the recently added base currency, `B` is the total base currency, `t` is the recently added token currency, `T` is the total token currency, and `P` is the current liquidity. The function attempts to convert the base and token currencies to liquidity on lines 11-15 through the following equation: $(P * (t * B) + (T * b)) / (T * B * 2)$, which is incorrect, with the correct equation being: $P * ((t * B) + (T * b)) / (T * B * 2)$. The understanding of the exact math is unnecessary, and therefore more details are excluded. As a consequence of this bug, all conversions of currencies to liquidity are incorrect, losing the funds of both the contract and the users. This bug was ranked as *High Risk* on Code4rena, the highest severity on the platform.

Example II (Trading Contract Tracer). Figure 2 depicts a function included within the *LibBalances* library from *Tracer* [13], which is a *derivative* smart contract designed to enable users to trade in *perpetual markets* [9]. A derivative contract has its functionalities based upon derivatives, namely, financial agreements derived from an underlying asset or financial market. Examples of derivatives include stocks, options, and futures. In perpetual markets, users can place long and short trade orders aiming on buying or selling *base* tokens using *quote* tokens, respectively. Users going long earn money when the price of the base token increases, while users going short earn money when the price of the base token decreases. Every user possesses a *position* that stores the amount of base and quote tokens that the user has. The *LibBalances* contract is a library within the *Tracer* project that provides basic functionalities in its perpetual market. In particular, the function `applyTrade()` is used to finalize both long and short trades. The amount of base token associated with the trade is defined as `signedAmount` on line 4, and the price of the base token is defined in the following line 5 as `signedPrice`. The amount and price of the base token are used to calculate the equivalent amount of quote token on line 6 as `quoteChange`. Then a fee for the transaction is calculated through the `getFee()` function call on line 7.

The short trade handling from line 14 to 16 is correct, and depicts the user selling the `signedAmount` in base token and earning the `quoteChange` in quote token, minus the fee by the contract. On the other hand, the long trade handling on lines 11 to 13 is incorrect,

specifically on line 13. Instead of losing the fee meant for the contract, the long trade user gains the fee instead. A likely explanation for this bug is that the developer confused the contract fee, which is charged to the user for each transaction, with the *funding fee* [8] in perpetual contracts, which is a fee paid from long traders to short traders periodically when the base token price increases, and from short traders to long traders when the base token price decreases, as an incentive for traders. The consequence of this bug is that users entering long trades would gain additional quote tokens on the contract's loss. This bug was also ranked as *High Risk* on Code4rena.

Existing Techniques Are Insufficient. As discussed in section 1, existing techniques such as static analysis, fuzzing, and symbolic execution require application agnostic oracles, like those used in finding reentrancy bugs. Reentrancy occurs when a victim contract makes a call to an external function before updating state variables, potentially allowing the external function to call the victim contract again. This leads to a loop which can cause unexpected behaviors such as fund draining. It can be detected by finding call cycles using various analysis: static, dynamic, and symbolic. Such an oracle does not have to model application specific behaviors. However, the previous two bugs reside in the core business logic specific to the contracts. For example, the first bug is coupled with the obscure math in function `calcLiquidityUnits()` and the second bug seems to require understanding perpetual trading and the unique design of position of the *Tracer* project. They hence cannot be detected by the automated tools we have tried (see section 4).

Our Technique. To address the aforementioned limitations, We propose a novel static analysis tool for detecting accounting bugs by considering inherent financial meanings of each variable. We introduce the concept of *extended type*, which denotes information such as token unit, scaling factor, and financial type for each variable and enables type propagation. This extended type information is used to check the correctness and consistency of all operations within the contract.

We demonstrate the usage of our tool here, starting with the bug in Figure 1 regarding the function `addLiquidity()`. Through type inference and propagation, `ScType` determines the token/currency units of `addedBase` on line 4, `addedToken` on line 5, and `totalLiquidity` on line 5, as t_{base} (meaning the token denoted by the address `base`), t_{token} , and $e(t_{base}, t_{token})$ denoting a composite expression involving t_{base} and t_{token} . More details of such inference can be found in subsection 3.4. In addition, due to line 8, `totalBase` has a token unit of t_{base} . Similarly, `totalToken` has a unit of t_{token} . On line 6 where the function `calcLiquidityUnits()` is invoked, our tool can infer that the parameters have token units of: $\{t_{base}, t_{base}, t_{token}, t_{token}, t_e = e(t_{base}, t_{token})\}$. This information is then propagated to the `calcLiquidityUnits()` function body starting on line 10. Hence on line 11, `part1 = t * B`, and therefore has unit $[t_{base} * t_{token}]$. On line 12, `part2 = T * b`, and has unit $[t_{base} * t_{token}]$. On line 13, `part3 = T * B * 2`, and therefore has unit $[t_{base} * t_{token}]$ as well. However, the issue arises on line 14 when the calculation of `_units = (((P * part1) + part2) / part3)` is performed. This calculation is split into three separate steps: the first calculation being `TMP_0 = P * part1`, the second calculation being `TMP_1 = TMP_0 + part2`, and finally, `_units = TMP_1 / part3`.

Project Name	Summary	A+	Func checked	TW	FP	TP	NTE	MTE	Analysis Time
MarginSwap	Dex project for margin trading on Uniswap and Sushiswap	6	20	1	0	1	1	0	8.93s
Vader Protocol	Yield project for a collateralized stablecoin	17	58	4	1	2	2	0	28.95s
PoolTogether	Gaming service on yield interest	15	8	1	0	1	1	0	8.94s
Tracer	Derivative project that supports perpetual markets	10	32	1	0	1	1	1	110.57s
Yield Micro	Lending project supporting borrowing, lending, and liquidity	4	18	2	0	1	1	1	10.39s
Sushi Trident	Dex project for deploying personalized liquidity markets	24	196	0	0	0	2	0	19.64s
yAxis	Yield project where users' aggregated funds are used in strategies for yield	9	59	4	1	2	1	1	16.81s
Badger Dao	Yield project	9	22	2	0	1	0	0	14.61s
Wild Credit	Lending project relying on pairs of assets instead of a pool	17	960	4	2	1	0	0	45.37s
PoolTogether v4	Gaming service on yield interest	1	23	0	0	0	1	0	14.99s
Sushi Trident p2	Dex project for deploying personalized liquidity markets	12	51	10	4	2	2	0	19.68s
Swivel	Yield project that allows users to set orders for Yield claiming	5	78	2	0	1	0	0	15.03s
Covalent	Users delegate commissions to a Validators, which stakes the funds for interest	6	59	1	1	0	0	1	15.87s
Badger Dao p2	Stablecoin to keep the internal contract asset price from fluctuating	6	14	2	0	1	0	0	10.31s
Vader Protocol	Yield project for a collateralized stablecoin (Swap)	54	169	13	2	6	2	0	66.86s
yAxis p2	Yield project where users' aggregated funds are used in strategies for yield	2	1	0	0	0	1	0	9.58s
Malt Finance	Stablecoin for the contract token, Malt	7	63	0	0	0	2	0	11.998s
Perennial	Derivative project supporting synthetic token perpetual markets	3	1	1	0	1	0	0	9.35s
Sublime	Lending contract dependent on trust minimization	15	57	5	2	2	0	0	24.54s
Yeti Finance	Lending project made against a contract specific token	9	34	0	0	0	1	0	12.35s
Vader Protocol p3	Yield project for a collateralized stablecoin	5	37	4	0	3	1	0	16.38s
InsureDao	Insurance markets where buyers pay premium for protection against losses	11	72	0	0	0	1	0	14.86s
Rocket Joe	Dex project where users exchange funds in return for new project liquidity	7	29	3	1	1	0	0	2.860s
Concur Finance	Yield project	4	41	0	0	0	1	0	2.980s
Biconomy Hyphen	Cross Chain project where users can deposit and withdraw for pools on different chains	6	60	1	0	1	0	0	14.64s
Sublime	Lending project allowing users to create custom lending pools	8	91	0	0	0	1	0	25.84s
Volt	Dex project which conserves the value of user funds against inflation	1	40	0	0	0	1	0	18.86s
Badger Dao p3	Yield project	8	68	0	0	0	1	0	16.22s
Tigris Trade	Dex project utilizing off-chain oracles to provide real-time prices	11	86	2	0	1	1	0	7.024s
Total				63	14	29	24	4	

* Annotations (A+), Total Warnings (TW), False Positive (FP), True Positive (TP), Not Type Error (NTE), Missed Type Error (MTE)

Table 3: Evaluation Results

Initial Type Annotations. Although ScType can automatically infer certain type information such as some token units and scaling factors, it may need the user to provide initial information such as financial types for some global variables and some function parameters (if they cannot be inferred). ScType prompts the user for such information when it is missing and cannot be resolved by type inference. Users' efforts are one-time and recorded in a type file for reuse. The information is clear from project description and code comments in most cases, and hence the required user efforts are limited, as demonstrated by the number of annotations in Table 3 (in comparison to the number of functions type-checked). More automation is certainly feasible. For example, financial types can be inferred from variable names in many cases or using mining techniques such as [46, 48]. We leave it to our future work.

The experiments are conducted on a machine with AMD Ryzen 3975x and 512GB RAM.

4.2 RQ1: Effectiveness

The results from running ScType on our benchmark are shown in Table 3. The projects are listed in the leftmost column, followed by a short summary of each project. The number of annotations made is shown in the column starting with A+. The number of functions checked is listed under the Func checked column, which includes functions that are called within other functions. The TW column indicates the number of type warnings by ScType. Singular true positives (TP) and false positives (FP) may both generate multiple warnings (due to the cascading effect of a type error), hence the discrepancy between the warnings amount and the sum of true and

```

1 int usdvReserve; //an amount of USDV
2 int vaderReserve; //an amount of VADER
3 function getRewardShare(address token) external {
4     uint baseAmount = getBaseAmount(token);
5     if (isVader(token)) {
6         uint _share = calcShare(_baseAmount,
7             ..., vaderReserve);
8         ...
9     } else if (isUSDV(token)) {
10        uint _share = calcShare(_baseAmount, ..., usdvReserve)
11        ;
12    }
13    ...
14 }

```

Figure 9: False Positive Example in Vader Protocol p1

false positive amounts. The true positives are listed under the TP column. Accounting bugs that are not type errors (and hence out of scope for ScType) are listed under the NTE, or Not-Type-Error column. Type bugs that are not able to be found with the current system are listed under the MTE, or Missed-Type-Error column.

Observations. In total, we run our tool on 29 projects, covering 57 accounting bugs. ScType reports 29 TPs and 14 FPs. The FPs are mainly due to the path insensitive nature of the tool. We will illustrate with a case later. Even though ScType cannot detect 28 of the 57 bugs, our inspection shows that 24 out of those 28 bugs are not type errors, belonging to other error categories such as pure math errors. Therefore, ScType is able to successfully detect $29/(29+4)=87.9\%$ of accounting type errors. We argue that these results demonstrate the promise of ScType as an attempt to addressing accounting bugs.

False positives. We manually inspect the false positives. We find that the lack of path sensitivity is a major reason. Figure 9 shows an example where ScType fails to type-check variable `baseAmount` within the function `getRewardShare()`, which takes as parameter an address `token` that represents the address of an arbitrary currency. Variable `baseAmount` is calculated as an amount of the token currency on line 5 via the function `getBaseAmount()`. Then, `token` is compared to the addresses of two currencies, VADER and USDV, by the function calls to `isVADER()` and `isUSDV()` on lines 5 and 9, respectively. If the `if` branch is taken on line 5, the function `getRewardShare()` swaps `baseAmount` of the VADER token. If not, USDV is swapped. ScType cannot resolve `baseAmount` to a unique token unit. Solving this problem may require path-sensitive analysis such as symbolic execution. We will leave this to our future work.

4.3 RQ2: Efficiency

To answer the research question regarding efficiency, we measure the cost of our tool, which is two-fold: the time of analysis and the total number of annotations. The former is shown in the last column of Table 3. Observe that the tool is very time affordable, with the maximum analysis time being less than 20 seconds. We point out that ScType automatically type-checks all functions that are being called, regardless of whether or not they reside in the same contract. The number of annotations is also reasonable given the large number of functions checked. Most of the annotations can be derived with minimum one-time manual efforts.

```

1 function belowMaintenanceThreshold(uint256 loan, uint256
   collateral) external {
2     ...
3     - return 100 *collateral >= liquidationThresholdPercent *loan;
4     + return 100 *collateral < liquidationThresholdPercent *loan;
5 }

```

Figure 10: Not-Type-Error Example in MarginSwap

```

1 uint256 totalReserve;
2 uint256 strategyReserve;
3 function setCap(uint256 cap) external {
4     ...
5     diff = strategyReserve - cap
6     - totalReserve -= strategyReserve;
7     + totalReserve -= diff;
8     ...
9 }

```

Figure 11: Missed-Type-Error Example in yAxis p1

4.4 RQ3: Distribution of Type Errors

We categorize the 33 different accounting type bugs (29 found by ScType and 4 missed) into 3 categories: token unit bugs, scaling factor bugs, and financial type bugs. Their numbers are 10, 12, and 11, respectively, which represent an approximately even distribution. The distribution strongly supports our current design. In addition, we find it a bit counter intuitive that token unit bugs are almost as common as the other two kinds, although they are simpler.

4.5 RQ4: Capacity of Type System

As shown in Table 3, ScType cannot detect 28 accounting bugs. Out of the 28, 24 are not type errors and deemed out-of-scope, while 4 are type errors that currently cannot be handled. In this section, we provide two case studies to illustrate these two types of bugs. Figure 10 shows a typical not-type-error (NTE). The bug lies in that the developers used the wrong comparison. Other NTE types include coefficient errors and even use of wrong formulas. These bugs need stronger oracles than type rules, such as input-output pairs and formal specification of business models. Figure 11 shows a missed-type-error (MTE). This function computes the amount that a certain reserve of some *strategy* (e.g., a contract that yields) exceeds a cap and removes the excess from the total reserve. In particular, the specific reserve is denoted by variable `strategyReserve`, the cap is `cap`, the excess is `diff` and the total reserve is `totalReserve`. The bug lies in that developers incorrectly subtract `strategyReserve`, while they should subtract `diff`. Although it is beyond our current system, a stronger type system that models balance delta such as `diff` as well as balance upper-bound like `cap` may prevent this bug. We leave this to our future work.

4.6 RQ5: Finding Zero-days

To study the real-world impact of ScType, we use it to audit a large real-world contract through Code4Rena. The project has over 10 thousands lines of code. We applied the technique on 9 contracts, found and reported 6 zero-days, with 4 of them leading to direct fund loss. Three of them are financial type errors and the other three are token unit bugs. We have created exploit inputs for these bugs as

proof-of-concepts. According to Code4Rena's policy, details should not be made public until the judges inspect all the bug reports and the developers are given the chances to fix the bugs.

5 THREATS TO VALIDITY

There is *internal* threat to validity due to human mistakes in type annotations. In practice, these annotations are mostly obvious from documentation and variable names. For example, fee and debt variables tend to have subwords "fee" and "debt" in their variable names. In addition, there may be implementation errors. As wrong annotations/implementations lead to spurious type errors, the few false positives by ScType indicate that the threat is mitigated. In the future, we plan to further reduce the human efforts (and hence the internal threat) by mining variables' financial meanings. The *external* threat mainly lies in the subjects used in our study. We mainly use the bugs in [78], which may not be representative. The risk is mitigated as all projects in the benchmark are real-world applications, with many having high complexity. The bug reports had gone through multiple rounds of interactions between auditors, developers, and Code4rena judges. In addition, we recently apply ScType to a very complex project and have encouraging results.

6 RELATED WORK

Detecting Business-related Vulnerabilities. Accounting bugs are related to business models. There have been pioneering efforts in detecting business related bugs. Wang et al. [69] proposed a fuzzing tool *Vultron*, which developed an interesting observation regarding *balance* and *transaction invariants*. In particular, the total balance of all the users and the contract should be the same, and transactions in or out of the contract should correspond to the same increase or decrease in total balance. This prevents bugs such as not updating a contract's balance after a withdraw. They later developed mining techniques to infer these invariants [46–48]. Fairness bug detection [49] aimed to detect unfair behaviors for game-like contracts having multiple participants. Sun et al. [44] developed a method to detect smart contract vulnerabilities based on a swap invariant and a transfer invariant, or logical rules that must be followed in order to facilitate proper functions. The technique can detect overflows and unprotected asset increases. In comparison, ScType can detect type problems that are largely complementary to the above works. Verification techniques [30, 32, 37, 38, 62, 64, 66, 72] are capable of detecting a wide spectrum of bugs including accounting bugs if the user can provide the specifications. In comparison, ScType encodes properties in its type rules.

Abstract Type Inference and Checking. ScType is essentially an abstract type system [16, 31, 33, 40–42, 50, 56, 59, 61] that derives abstract types with much richer semantics than primitive types. However, existing techniques do not focus on smart contracts, which have very unique finance oriented semantics. Tan et al. [64] developed a refinement type system known as *SolType* for Solidity. It models low level relationships between integers and checks for overflows/underflows.

Smart Contract Bug Finding. ScType is related to smart contract bug finding in general, including static analysis [18, 23, 26, 27, 35, 39, 60, 65, 68, 71, 79], fuzzing [24, 28, 29, 34, 36, 45, 54, 67, 69, 70, 74, 75],

and symbolic execution [1, 17, 19, 43, 51, 52, 55, 61, 67, 73]. In contract, ScType focuses finding accounting bugs, complementary to these techniques.

Bug Studies. We are inspired by a recent study on smart contract vulnerabilities [78], which showed the prevalence of accounting bugs and the difficulty of finding them, and also by a list of other comprehensive studies of various kinds of smart contract bugs and programming practices [20–22, 25, 57, 58, 76, 77].

7 CONCLUSION

We develop an abstract type inference and checking technique to detect accounting bugs in smart contracts, a kind of bug difficult for existing automatic tools. The technique models token units, scaling factors, and financial meanings of individual variables and checks type consistencies. Our results show that more 58% of known accounting bugs are type errors, and our tool detects 87.9% of these type errors. It also finds 6 zero-days.

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